Naoto Ohsaka

Curriculum vitae

Personal and Contact Information

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Education

March 2018 **Doctor of Information Science and Technology**
Graduate School of Information Science and Technology, the University of Tokyo, Japan
Title: Efficient and Effective Identification of Influential Vertices in Social Networks

March 2015 **Master of Information Science and Technology**
Graduate School of Information Science and Technology, the University of Tokyo, Japan
Title: Estimating and Maximizing the Spread of Influence in Social Networks: Pruned Monte-Carlo Simulations and Fully-Dynamic Indices

March 2013 **Bachelor of Engineering**
Department of Computer Science, the University of Electro-Communications, Japan
Title: Study on Improving the Performance of a Streaming Algorithm for the k-means Problem

Employment

April 2013–March 2016	Research Assistant, *JST ERATO Kawarabayashi Large Graph Project*
April 2016–March 2018	Research Fellow, *Japan Society for the Promotion of Science (DC2)*
April 2018–March 2020	Researcher, *NEC Corporation*
April 2020–November 2021	Senior Researcher, *NEC Corporation*
December 2021–Present	Research Scientist, *CyberAgent, Inc.*

Journal Articles & Refereed Conference Proceedings

In theoretical computer science (e.g., STOC, FOCS, SODA, and ICALP), refereed conference proceedings are the primary publication venues, and the authors are listed in alphabetical order. For other papers, the authors are listed in order of contribution.

1. Naoto Ohsaka, Daisuke Kitakoshi, and Masato Suzuki. "A Reinforcement Learning Method to Improve the Sweeping Efficiency for an Agent". *Proceedings of the 2011 IEEE International Conference on Granular Computing (**GrC**)*, pp. 515–520, 2011. DOI: https://doi.org/10.1109/GRC.2011.6122650
2. Naoto Ohsaka, Takuya Akiba, Yuichi Yoshida, and Ken-ichi Kawarabayashi. "Fast and Accurate Influence Maximization on Large Networks with Pruned Monte-Carlo Simulations". *Proceedings of the 28th AAAI Conference on Artificial Intelligence (**AAAI**)*, pp. 138–144, 2014. DOI: https://doi.org/10.1609/aaai.v28i1.8726
3. Naoto Ohsaka, Takanori Maehara, and Ken-ichi Kawarabayashi. "Efficient PageRank Tracking in Evolving Networks". *Proceedings of the 21st ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (**KDD**)*, pp. 875–884, 2015. DOI: https://doi.org/10.1145/2783258.2783297
4. Naoto Ohsaka and Yuichi Yoshida. "Monotone *k*-Submodular Function Maximization with Size Constraints". *Proceedings of the 29th Annual Conference on Neural Information Processing Systems (**NIPS**)*, pp. 694–702, 2015. URL: http://papers.nips.cc/paper/5709-monotone-k-submodular-function-maximization-with-size-constraints
5. Naoto Ohsaka, Takuya Akiba, Yuichi Yoshida, and Ken-ichi Kawarabayashi. "Dynamic Influence Analysis in Evolving Networks". *Proceedings of the VLDB Endowment, (**PVLDB**)*, 9(12), pp. 1077–1088, 2016. URL: http://www.vldb.org/pvldb/vol9/p1077-ohsaka.pdf
6. Naoto Ohsaka, Yutaro Yamaguchi, Naonori Kakimura, and Ken-ichi Kawarabayashi. "Maximizing Time-Decaying Influence in Social Networks". *Proceedings of the 15th European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (**ECML PKDD**)*, pp. 132–147, 2016. DOI: https://doi.org/10.1007/978-3-319-46128-1_9
7. Naoto Ohsaka and Yuichi Yoshida. "Portfolio Optimization for Influence Spread". *Proceedings of the 26th International Conference on World Wide Web (**WWW**)*, pp. 977–985, 2017. DOI: https://doi.org/10.1145/3038912.3052628
8. Naoto Ohsaka, Tomohiro Sonobe, Sumio Fujita, and Ken-ichi Kawarabayashi. "Coarsening Massive Influence Networks for Scalable Diffusion Analysis". *Proceedings of the 2017 ACM SIGMOD International Conference on Management of Data (**SIGMOD**)*, pp. 635–650, 2017. DOI: https://doi.org/10.1145/3035918.3064045
9. Yoichi Iwata, Tomoaki Ogasawara, and Naoto Ohsaka. "On the Power of Tree-Depth for Fully Polynomial FPT Algorithms". *Proceedings of the 35th International Symposium on Theoretical Aspects of Computer Science (**STACS**)*, pp. 41:1–41:14, 2018. DOI: https://doi.org/10.4230/LIPIcs.STACS.2018.41
10. Diana Popova, Naoto Ohsaka, Ken-ichi Kawarabayashi, and Alex Thomo. "NoSingles: A Space-Efficient Algorithm for Influence Maximization". *Proceedings of the 30th International Conference on Scientific and Statistical Database Management (**SSDBM**)*, pp. 18:1–18:12, 2018. DOI: https://doi.org/10.1145/3221269.3221291
11. Naoto Ohsaka, Tomohiro Sonobe, Naonori Kakimura, Takuro Fukunaga, Sumio Fujita, and Ken-ichi Kawarabayashi. "Boosting PageRank Scores by Optimizing Internal Link Structure". *Proceedings of the 29th*

*International Conference on Database and Expert Systems Applications (**DEXA**)*, pp. 424–439, 2018. DOI: https://doi.org/10.1007/978-3-319-98809-2_26

12. Naoto Ohsaka, Tomoya Sakai, and Akihiro Yabe. "A Predictive Optimization Framework for Hierarchical Demand Matching". *Proceedings of the 2020 SIAM International Conference on Data Mining (**SDM**)*, pp. 172–180, 2020. DOI: https://doi.org/10.1137/1.9781611976236.20
13. Naoto Ohsaka. "The Solution Distribution of Influence Maximization: A High-level Experimental Study on Three Algorithmic Approaches". *Proceedings of the 2020 ACM SIGMOD International Conference on Management of Data (**SIGMOD**)*, pp. 2151–2166, 2020. DOI: https://doi.org/10.1145/3318464.3380564
14. Naoto Ohsaka and Tatsuya Matsuoka. "On the (In)tractability of Computing Normalizing Constants for the Product of Determinantal Point Processes". *Proceedings of the 37th International Conference on Machine Learning (**ICML**)*, pp. 7414–7423, 2020. URL: http://proceedings.mlr.press/v119/ohsaka20a.html
15. Tomoya Sakai and Naoto Ohsaka. "Predictive Optimization with Zero-Shot Domain Adaptation". *Proceedings of the 2021 SIAM International Conference on Data Mining (**SDM**)*, pp. 369–377, 2021. DOI: https://doi.org/10.1137/1.9781611976700.42
16. Naoto Ohsaka. "Unconstrained MAP Inference, Exponentiated Determinantal Point Processes, and Exponential Inapproximability". *Proceedings of the 24th International Conference on Artificial Intelligence and Statistics (**AISTATS**)*, pp. 154–162, 2021. URL: https://proceedings.mlr.press/v130/ohsaka21a.html
17. Tatsuya Matsuoka, Shinji Ito, and Naoto Ohsaka. "Tracking Regret Bounds for Online Submodular Optimization". *Proceedings of the 24th International Conference on Artificial Intelligence and Statistics (**AISTATS**)*, pp. 3421–3429, 2021. URL: https://proceedings.mlr.press/v130/matsuoka21a.html
18. Tatsuya Matsuoka and Naoto Ohsaka. "Spanning Tree Constrained Determinantal Point Processes are Hard to (Approximately) Evaluate". *Operations Research Letters (**ORL**)*, 49(3), pp. 304–309, 2021. DOI: https://doi.org/10.1016/j.orl.2021.02.004
19. Naoto Ohsaka. "A Fully Polynomial Parameterized Algorithm for Counting the Number of Reachable Vertices in a Digraph". *Information Processing Letters (**IPL**)*, 171, pp. 106137, 2021. DOI: https://doi.org/10.1016/j.ipl.2021.106137
20. Naoto Ohsaka and Tatsuya Matsuoka. "Approximation Algorithm for Submodular Maximization under Submodular Cover". *Proceedings of the 37th Conference on Uncertainty in Artificial Intelligence (**UAI**)*, pp. 792–801, 2021. URL: https://proceedings.mlr.press/v161/ohsaka21a.html
21. Tatsuya Matsuoka, Naoto Ohsaka, and Akihiro Yabe. "On the Convex Combination of Determinantal Point Processes". *Proceedings of the 13th Asian Conference on Machine Learning (**ACML**)*, pp. 158–173, 2021. URL: https://proceedings.mlr.press/v157/matsuoka21a.html
22. Tatsuya Matsuoka and Naoto Ohsaka. "Maximization of Monotone k-Submodular Functions with Bounded Curvature and Non-k-Submodular Functions". *Proceedings of the 13th Asian Conference on Machine Learning (**ACML**)*, pp. 1707–1722, 2021. URL: https://proceedings.mlr.press/v157/matsuoka21b.html

23. Naoto Ohsaka and Tatsuya Matsuoka. "Reconfiguration Problems on Submodular Functions". *Proceedings of the 15th ACM International Conference on Web Search and Data Mining (**WSDM**)*, pp. 764–774, 2022. DOI: https://doi.org/10.1145/3488560.3498382
24. Naoto Ohsaka. "Some Inapproximability Results of MAP Inference and Exponentiated Determinantal Point Processes". *Journal of Artificial Intelligence Research (**JAIR**)*, 73, pp. 709–735, 2022. DOI: https://doi.org/10.1613/jair.1.13288
25. Naoto Ohsaka. "On the Parameterized Intractability of Determinant Maximization". *Proceedings of the 33rd International Symposium on Algorithms and Computation (**ISAAC**)*, pp. 46:1–46:16, 2022. DOI: https://doi.org/10.4230/LIPIcs.ISAAC.2022.46
26. Naoto Ohsaka. "On Reconfigurability of Target Sets". *Theoretical Computer Science (**TCS**)*, 942, pp. 253–275, 2023. DOI: https://doi.org/10.1016/j.tcs.2022.11.036
27. Naoto Ohsaka. "Gap Preserving Reductions Between Reconfiguration Problems". *Proceedings of the 40th International Symposium on Theoretical Aspects of Computer Science (**STACS**)*, pp. 49:1–49:18, 2023. DOI: https://doi.org/10.4230/LIPIcs.STACS.2023.49
28. Naoto Ohsaka and Riku Togashi. "A Critical Reexamination of Intra-List Distance and Dispersion". *Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval (**SIGIR**)*, pp. 1619–1628, 2023. DOI: https://doi.org/10.1145/3539618.3591623
29. Naoto Ohsaka and Riku Togashi. (equal contribution) "Curse of "Low" Dimensionality in Recommender Systems". *Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval (**SIGIR**)*, pp. 537–547, 2023. DOI: https://doi.org/10.1145/3539618.3591659
30. Yoji Tomita, Riku Togashi, Yuriko Hashizume, and Naoto Ohsaka. "Fast and Examination-agnostic Reciprocal Recommendation in Matching Markets". *Proceedings of the 17th ACM Conference on Recommender Systems (**RecSys**)*, pp. 12–23, 2023. DOI: https://doi.org/10.1145/3604915.3608774
31. Naoto Ohsaka. "Gap Amplification for Reconfiguration Problems". *Proceedings of the 35th Annual ACM-SIAM Symposium on Discrete Algorithms (**SODA**)*, pp. 1345–1366, 2024. DOI: https://doi.org/10.1137/1.9781611977912.54
32. Naoto Ohsaka. "On the Parameterized Intractability of Determinant Maximization". *Algorithmica (Special issue of **ISAAC** 2022)*, 86(6), pp. 1731–1763, 2024. DOI: https://doi.org/10.1007/s00453-023-01205-0
33. Riku Togashi, Tatsushi Oka, Naoto Ohsaka, and Tetsuro Morimura. "Safe Collaborative Filtering". *Proceedings of the 12th International Conference on Learning Representations (**ICLR**)*, 2024. URL: https://openreview.net/pdf?id=yarUvgEXq3
34. Shuichi Hirahara and Naoto Ohsaka. "Probabilistically Checkable Reconfiguration Proofs and Inapproximability of Reconfiguration Problems". *Proceedings of the 56th Annual ACM Symposium on Theory of Computing (**STOC**)*, pp. 1435–1445, 2024. DOI: https://doi.org/10.1145/3618260.3649667

35. Tatsuya Matsuoka and Naoto Ohsaka. "Computational Complexity of Normalizing Constants for the Product of Determinantal Point Processes". *Theoretical Computer Science (**TCS**)*, 997, pp. 114513, 2024. DOI: https://doi.org/10.1016/j.tcs.2024.114513
36. Shuichi Hirahara and Naoto Ohsaka. "Optimal PSPACE-hardness of Approximating Set Cover Reconfiguration". *Proceedings of the 51st EATCS International Colloquium on Automata, Languages, and Programming (**ICALP**)*, pp. 85:1–85:18, 2024. DOI: https://doi.org/10.4230/LIPIcs.ICALP.2024.85
37. Naoto Ohsaka. "Alphabet Reduction for Reconfiguration Problems". *Proceedings of the 51st EATCS International Colloquium on Automata, Languages, and Programming (**ICALP**)*, pp. 113:1–113:17, 2024. DOI: https://doi.org/10.4230/LIPIcs.ICALP.2024.113
38. Ruo-Chun Tzeng, Naoto Ohsaka, and Kaito Ariu. "Matroid Semi-Bandits in Sublinear Time". *Proceedings of the 40th International Conference on Machine Learning (**ICML**)*, 2024. URL: https://openreview.net/forum?id=MwQ53xAIPs
39. Naoto Ohsaka. "On Approximate Reconfigurability of Label Cover". *Information Processing Letters (**IPL**)*, 189, pp. 106556, 2025. DOI: https://doi.org/10.1016/j.ipl.2024.106556
40. Naoto Ohsaka. "Yet Another Simple Proof of the PCRP Theorem". *Proceedings of the 52nd EATCS International Colloquium on Automata, Languages, and Programming (**ICALP**)*, pp. 122:1–122:18, 2025. DOI: https://doi.org/10.4230/LIPIcs.ICALP.2025.122
41. Shuichi Hirahara and Naoto Ohsaka. "Asymptotically Optimal Inapproximability of Maxmin *k*-Cut Reconfiguration". *Proceedings of the 52nd EATCS International Colloquium on Automata, Languages, and Programming (**ICALP**)*, pp. 96:1–96:18, 2025. DOI: https://doi.org/10.4230/LIPIcs.ICALP.2025.96
42. Shuichi Hirahara and Naoto Ohsaka. "Asymptotically Optimal Inapproximability of E*k*-SAT Reconfiguration". *Proceedings of the 66th IEEE Symposium on Foundations of Computer Science (**FOCS**)*, to appear, 2025.
43. Shuichi Hirahara, Naoto Ohsaka, Tatsuhiro Suga, Akira Suzuki, Yuma Tamura, and Xiao Zhou. "Reachability of Independent Sets and Vertex Covers Under Extended Reconfiguration Rules". *Proceedings of the 36th International Symposium on Algorithms and Computation (**ISAAC**)*, to appear, 2025.
44. Naoto Ohsaka. "Gap Preserving Reductions Between Reconfiguration Problems". *Journal of Computer and System Sciences (**JCSS**) (Full version of **STACS** 2023)*, to appear, 2025.

Invited Talks

1. 20th CoRe Seminar, KAKENHI Grant-in-Aid for Transformative Research Areas (B), Online, January 6th, 2022. "On Reconfigurability of Target Sets"
2. Kawarabayashi Lab Seminar, the University of Tokyo, Japan, January 19th, 2024. "Reconfiguration Problems, Hardness of Approximation, and Gap Amplification: What Are They?"
3. Theoretical Foundations of Computing (COMP), Tokyo, Japan, March 14th, 2024. "Gap Amplification for Reconfiguration Problems"

4. The 5th Combinatorial Reconfiguration Workshop, Fukuoka, Japan, October 8th, 2024.
   "On the Complexity of Approximating Reconfiguration Problems"

Refereeing

- Conferences: AAAI 2016, NeurIPS 2019, ICML 2020, NeurIPS 2020, AAAI 2021, AISTATS 2021, ICML 2021, ICML 2022, MFCS 2023, ISAAC 2023, CCC 2024, ICALP 2025, ESA 2025, APPROX 2025
- Journals: IEICE Transactions on Information and Systems (2015; 2018; 2023), IEEE Access (2019), Algorithmica (2019), PLOS ONE (2020), The VLDB Journal (2021), ACM Transactions on Database Systems (2022), Theory of Computing (2025)

Patents

1. Manao Machida and Naoto Ohsaka. Route search support apparatus, route search support method, and computer-readable recording medium. US20220019235A1, issued January 20th, 2022.
2. Akihiro Yabe and Naoto Ohsaka. Information processing device, control method, and storage medium. US20220261822A1, issued August 18th, 2022.
3. Tatsuya Matsuoka, Naoto Ohsaka, and Akihiro Yabe. Information processing device, control method and storage medium. US20230118145A1, issued April 20th, 2023.
4. Naoto Ohsaka and Tatsuya Matsuoka. Information processing device, control method, and storage medium. US20230245188A1, issued August 3rd, 2023.
5. Tatsuya Matsuoka and Naoto Ohsaka. Information processing device, control method and storage medium. US20230325463A1, issued October 12th, 2023.
6. Shinji Ito, Tatsuya Matsuoka, and Naoto Ohsaka. Information processing device. US20240020351A1, issued January 18th, 2024.
7. Naoto Ohsaka and Tatsuya Matsuoka. Information processing device, control method, and storage medium. US20240086985A1, issued March 14th, 2024.
8. Naoto Ohsaka and Tatsuya Matsuoka. Information processing device, control method, and storage medium. US20250005463A1, issued January 2nd, 2025.
9. Naoto Ohsaka and Tatsuya Matsuoka. Information processing device, control method, and storage medium. US20250013935A1, issued January 9th, 2025.
10. Tatsuya Matsuoka and Naoto Ohsaka. Learning device, presentation device, learning method, and storage medium. US20250103672A1, issued March 27th, 2025.

Awards and Honors

November 2012 — 3rd Place (with Izuru Matsuura and Masafumi Yabu), ACM International Collegiate Programming Contest Asia Regional Contest 2012 in Tokyo, Tokyo, Japan

July 2013	14th Place (with Izuru Matsuura and Masafumi Yabu), ACM International Collegiate Programming Contest World Finals 2013, St. Petersburg, Russia
March 2021	DBSJ Kambayashi Young Researcher Award, The Database Society of Japan